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ME 5130: Advanced Heat and Mass Transfer
Midterm Exam: Take-Home
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Problem 2: Plotting of the Temperature Solution

Consider the assigned homework problem 2-15 from Ozisik (1993). Use $a = 1$ m, $b = 1$ m, $k = 0.8$ W/m-K, $h = 10$ W/m^2-K, $T = f(x) = 100$ K.

First, the solution of Özışık (1993) Problem 2–15, derived in Homework 01, is the steady-state temperature distribution $T(x, y)$ for a rectangular region $0 \leq x \leq a$, $0 \leq y \leq b$ that is insulated at $x = 0$, held at $T = f(x)$ on $y = 0$, and dissipates heat by convection to an environment at zero temperature through $x = a$ and $y = b$. After defining $H = h/k$, the problem can be expressed as:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0, \qquad 0 < x < a,\ 0 < y < b \tag{1}$$

$$\frac{\partial T}{\partial x} = 0, \qquad x = 0 \tag{2}$$

$$\frac{\partial T}{\partial x} + HT = 0, \qquad x = a \tag{3}$$

$$T(x, 0) = f(x), \qquad y = 0 \tag{4}$$

$$\frac{\partial T}{\partial y} + HT = 0, \qquad y = b \tag{5}$$

After separating variables and obtaining the eigenfunction and normalization factor from Table 2–2 (Özışık (1993)), the following results were obtained for Homework 01:

$$T(x, y) = \sum_{m=1}^{\infty} \frac{\cos \beta_m x \left[\beta_m \cosh \beta_m (b - y) + H \sinh \beta_m (b - y)\right]}{N(\beta_m) \left[\beta_m \cosh \beta_m b + H \sinh \beta_m b\right]} \int_0^a \cos \beta_m x' \, f(x') \, dx' \tag{6}$$

where:

$$\frac{1}{N(\beta_m)} = \frac{2\left(\beta_m^2 + H^2\right)}{a\left(\beta_m^2 + H^2\right) + H} \tag{7}$$

and the eigenvalues β_m are the positive roots of:

$$\beta_m \tan \beta_m a = H \tag{8}$$

For this problem, $f(x) = T_0 = 100$ K is a constant, therefore:

$$\int_0^a \cos\beta_m x'\, T_0\, dx' = T_0 \left[\frac{\sin\beta_m x'}{\beta_m}\right]_0^a = \frac{T_0 \sin\beta_m a}{\beta_m} \tag{9}$$

Substituting this into the Homework 01 solution above gives the expression evaluated in the code:

$$T(x,y) = \sum_{m=1}^{\infty} \frac{T_0 \sin\beta_m a}{\beta_m N(\beta_m)} \cos\beta_m x\, \frac{\beta_m \cosh\beta_m(b-y) + H\sinh\beta_m(b-y)}{\beta_m \cosh\beta_m b + H\sinh\beta_m b} \tag{10}$$

2a.

List the first ten eigenvalues β_m used to generate your contour plot.

Problem 2a: Solution

Table 1: First ten eigenvalues β_m, the positive roots of $\beta_m \tan\beta_m a = H$, for $a = 1$ m and $H = h/k = 12.5$ m^{-1}.

m	β_m (m^{-1})
1	1.4549
2	4.3757
3	7.3240
4	10.3061
5	13.3200
6	16.3604
7	19.4214
8	22.4983
9	25.5872
10	28.6853

2b.

Create a 2-D contour plot of the temperature field $T(x,y)$ for $[\,0 \le x \le a;\ 0 \le y \le b\,]$. Show at least 6 isotherms, and numerically label the temperature value of all the isotherms.

Problem 2b: Solution

Figure 1 shows nine isotherms, from 10 K to 90 K. The isotherms and contour plot show that heat enters the plate through the bottom face at 100 K and leaves by convection through the top and right faces. The isotherms meet the insulated face $x = 0$ at right angles because $\partial T/\partial x = 0$ at the face (and the temperature gradient is perpendicular to the isotherms). The isotherms also cluster together near the corner $(a, 0)$, where the 100 K face meets a cooled face and the temperature gradient is large.

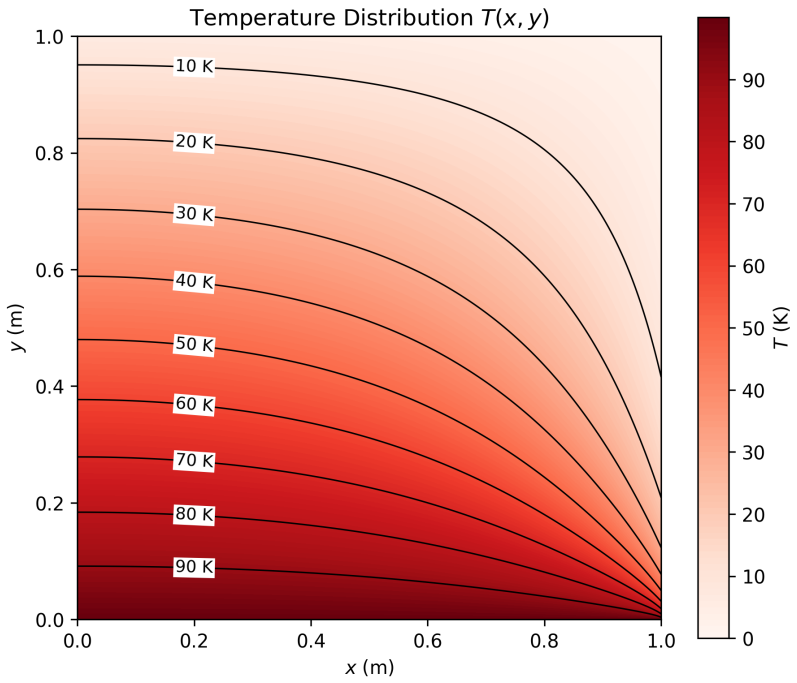


Figure 1: Isotherms of the steady temperature distribution $T(x, y)$ for $a = b = 1$ m, $k = 0.8$ W/m-K, $h = 10$ W/m^2-K and $f(x) = 100$ K, with the series summed to 1000 terms.

Problem 3: Diffusion Mass Transfer

For this problem, consider example problem 17.4-1 in Bird, Stewart and Lightfoot (1960), pg. 534–535.

3a.

Using the Shell Mass Balance approach:

$$[\text{in}] - [\text{out}] + [\text{reaction}] = 0$$

create a sketch of a control volume, and indicate all the in/out fluxes, and reaction terms in your sketch.

Problem 3a: Solution

Using the shell mass balance approach, we can evaluate a mass balance on a thin shell perpendicular to the direction of A's motion. Given that A moves in the z-direction, and assuming δ is small compared with the bubble diameter, let the shell be a thin slab between z and $z + \Delta z$ with surface area S and volume $S\Delta z$. We can now evaluate the following shell balance:

$$[\text{rate of } A \text{ in}] - [\text{rate of } A \text{ out}] + \begin{bmatrix}\text{rate of production of } A \\ \text{by homogeneous} \\ \text{chemical reaction}\end{bmatrix} = 0 \tag{11}$$

Where the rate of A in is expressed as:

$$\text{Rate in} = N_{Az}\big|_z \, S \tag{12}$$

The rate of A out is expressed as:

$$\text{Rate out} = N_{Az}\big|_{z+\Delta z} \, S \tag{13}$$

and the rate of production of A by a homogeneous first-order chemical reaction is:

$$\text{reaction} = -k_1''' c_A \, S \, \Delta z \tag{14}$$

Therefore, the shell balance can be expressed as:

$$N_{Az}\big|_z \, S - N_{Az}\big|_{z+\Delta z} \, S - k_1''' c_A \, S \, \Delta z = 0 \tag{15}$$

and represented visually as:

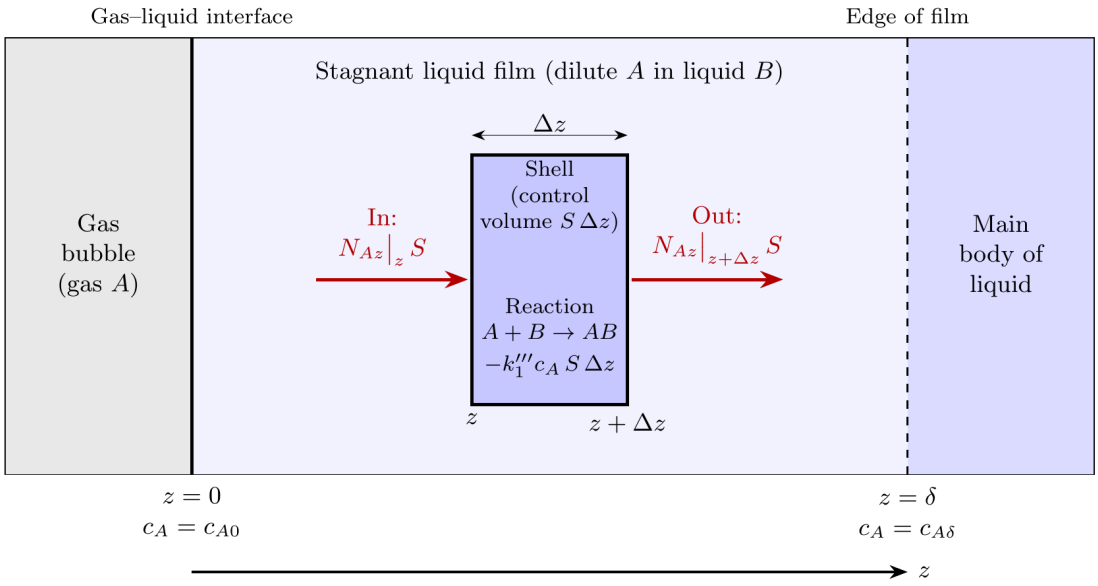


Figure 2: Shell mass balance on gas A over a slab of thickness Δz in the stagnant liquid film.

3b.

Use this Shell Mass Balance sketch to derive the following governing equation and boundary conditions.

Governing equation:

$$-\mathcal{D}_{AB}\frac{d^2c_A}{dz^2} + k_1'''c_A = 0$$

Boundary conditions:

$$c_A = c_{A0} \quad \text{at } z = 0$$

$$c_A = c_{A\delta} \quad \text{at } z = \delta$$

All assumptions must be clearly stated, and all steps clearly explained.

Problem 3b: Solution

The shell balance above can now be used to derive the governing equation and boundary conditions for the problem under the following assumptions:

1. A steady-state concentration is quickly established in the liquid film after the bubble is formed (BSL 1960 p. 534). Therefore, c_A in the film does not change with time.
2. Each gas bubble is surrounded by a stagnant liquid film of thickness δ, which is small with respect to the bubble diameter (BSL 1960 p. 534).
3. The gas A is only sparingly soluble in the liquid, so that we can neglect the bulk flow term in BSL Equation 17.0-1 (BSL 1960 p. 534).
4. The bulk of the liquid outside the stagnant film is at concentration $c_{A\delta}$, which changes so slowly with respect to time that it can be considered constant (BSL 1960 p. 534).
5. The total molar concentration c is virtually uniform throughout the liquid, and therefore constant.
6. Constant properties throughout the stagnant liquid film, therefore $\mathcal{D}_{AB}$ and k_1''' are constant.
7. The reaction $A + B \rightarrow AB$ is a homogeneous, irreversible first-order reaction (BSL 1960 p. 534).
8. The product of the chemical reaction AB does not interfere with the diffusion of A through B.
9. c_{A0} is the interfacial concentration of A in the liquid phase and is at equilibrium with the gas phase at the interface (BSL 1960 p. 534-535).

Starting with:

$$N_{Az}\big|_z S - N_{Az}\big|_{z+\Delta z} S - k_1'''c_A S\,\Delta z = 0 \tag{16}$$

The equation above can be simplified to:

$$\begin{aligned} \frac{N_{Az}\big|_z - N_{Az}\big|_{z+\Delta z}}{\Delta z} - k_1''' c_A = 0 \\ -\frac{N_{Az}\big|_{z+\Delta z} - N_{Az}\big|_z}{\Delta z} - k_1''' c_A = 0 \end{aligned} \tag{17}$$

Taking the limit as $\Delta z \to 0$ yields:

$$\lim_{\Delta z \to 0} \frac{N_{Az}\big|_{z+\Delta z} - N_{Az}\big|_z}{\Delta z} = \frac{dN_{Az}}{dz} \tag{18}$$

Therefore:

$$\begin{aligned} -\frac{dN_{Az}}{dz} - k_1''' c_A = 0 \\ \frac{dN_{Az}}{dz} + k_1''' c_A = 0 \end{aligned} \tag{19}$$

Now, Equation 17.0-1 from BSL writes:

$$N_{Az} = -c\,\mathcal{D}_{AB}\frac{dx_A}{dz} + x_A\,(N_{Az} + N_{Bz}) \tag{20}$$

However, since it is assumed that the gas A is only sparingly soluble in the liquid, we can neglect the bulk flow term, $x_A\,(N_{Az} + N_{Bz})$. Thus:

$$N_{Az} = -c\,\mathcal{D}_{AB}\frac{dx_A}{dz} \tag{21}$$

And because c is constant, and $c_A = cx_A$, the equation above can be written as:

$$N_{Az} = -\mathcal{D}_{AB}\frac{dc_A}{dz} \tag{22}$$

The equation above can now be inserted into the differential equation obtained from the shell balance:

$$\frac{d}{dz}\left(-\mathcal{D}_{AB}\frac{dc_A}{dz}\right) + k_1''' c_A = 0 \tag{23}$$

and since $\mathcal{D}_{AB}$ is assumed to be constant:

$$\boxed{-\mathcal{D}_{AB}\frac{d^2 c_A}{dz^2} + k_1''' c_A = 0} \tag{24}$$

Which can be classified as a second-order differential equation that requires two conditions for a particular solution. At the interface $z = 0$, it is assumed that c_{A0} is the interfacial concentration of A in the liquid phase and is at equilibrium with the gas phase. Therefore:

$$\boxed{c_A = c_{A0} \quad \text{at } z = 0} \tag{25}$$

At the edge of the film ($z = \delta$), the bulk of the liquid outside the stagnant film is assumed to be at concentration $c_{A\delta}$, which changes so slowly with respect to time that it can be considered constant (BSL 1960 p. 534). Therefore:

$$\boxed{c_A = c_{A\delta} \quad \text{at } z = \delta} \tag{26}$$

3c.

Show that the solution to this system is the following. All assumptions must be clearly stated, and all steps clearly explained.

$$\frac{c_A}{c_{A0}} = \frac{\Gamma \sinh b_1 \zeta + \sinh b_1 (1 - \zeta)}{\sinh b_1}$$

Problem 3c: Solution

Maintaining the same assumptions as above and adding the following:

1. $k_1''' > 0$, $\delta > 0$, and $\mathcal{D}_{AB} > 0$, so $b_1 > 0$ and $\sinh b_1 > 0$.
2. $c_{A0} > 0$.

The governing equation above can be considered:

$$\begin{aligned} -\mathcal{D}_{AB} \frac{d^2 c_A}{dz^2} + k_1''' c_A &= 0 \\ \frac{d^2 c_A}{dz^2} - \frac{k_1'''}{\mathcal{D}_{AB}} c_A &= 0 \end{aligned} \tag{27}$$

Let:

$$\lambda^2 = \frac{k_1'''}{\mathcal{D}_{AB}} \tag{28}$$

Therefore:

$$\frac{d^2 c_A}{dz^2} - \lambda^2 c_A = 0 \tag{29}$$

A second-order homogeneous linear ordinary differential equation with constant coefficients and a well-known general solution:

$$c_A = C_1 \cosh \lambda z + C_2 \sinh \lambda z \tag{30}$$

The boundary conditions above can now be applied. At $z = 0$, $c_A = c_{A0}$:

$$\begin{aligned} c_{A0} &= C_1 \cosh 0 + C_2 \sinh 0 \\ c_{A0} &= C_1 \end{aligned} \tag{31}$$

At $z = \delta$, $c_A = c_{A\delta}$:

$$
\begin{aligned}
c_{A\delta} &= c_{A0}\cosh\lambda\delta + C_2\sinh\lambda\delta \\
C_2 &= \frac{c_{A\delta} - c_{A0}\cosh\lambda\delta}{\sinh\lambda\delta}
\end{aligned}
\tag{32}
$$

Therefore:

$$
c_A = c_{A0}\cosh\lambda z + \frac{c_{A\delta} - c_{A0}\cosh\lambda\delta}{\sinh\lambda\delta}\,\sinh\lambda z \tag{33}
$$

Given that $\Gamma = \dfrac{c_{A\delta}}{c_{A0}}$, dividing both sides by c_{A0} gives:

$$
\frac{c_A}{c_{A0}} = \cosh\lambda z + \frac{\Gamma - \cosh\lambda\delta}{\sinh\lambda\delta}\,\sinh\lambda z \tag{34}
$$

Which can be rearranged to:

$$
\begin{aligned}
\frac{c_A}{c_{A0}} &= \frac{\sinh\lambda\delta\cosh\lambda z + \Gamma\sinh\lambda z - \cosh\lambda\delta\sinh\lambda z}{\sinh\lambda\delta} \\
&= \frac{\Gamma\sinh\lambda z + (\sinh\lambda\delta\cosh\lambda z - \cosh\lambda\delta\sinh\lambda z)}{\sinh\lambda\delta}
\end{aligned}
\tag{35}
$$

and using the identity: $\sinh(A-B) = \sinh A\cosh B - \cosh A\sinh B$:

$$
\frac{c_A}{c_{A0}} = \frac{\Gamma\sinh\lambda z + \sinh\lambda(\delta - z)}{\sinh\lambda\delta} \tag{36}
$$

Now, given:

$$
\zeta = \frac{z}{\delta}, \qquad b_1 = \sqrt{\frac{k_1'''\delta^2}{\mathcal{D}_{AB}}} \tag{37}
$$

and recalling that:

$$
\lambda = \sqrt{\frac{k_1'''}{\mathcal{D}_{AB}}} \tag{38}
$$

Therefore, since $\delta > 0$:

$$
\lambda\delta = \sqrt{\frac{k_1'''}{\mathcal{D}_{AB}}}\,\delta = \sqrt{\frac{k_1'''\,\delta^2}{\mathcal{D}_{AB}}} = b_1 \tag{39}
$$

$$
\lambda z = \lambda\delta\,\frac{z}{\delta} = b_1\zeta \tag{40}
$$

$$
\lambda(\delta - z) = \lambda\delta\left(1 - \frac{z}{\delta}\right) = b_1(1-\zeta) \tag{41}
$$

and the normalized molar concentration of gas A can be expressed as:

$$
\boxed{\frac{c_A}{c_{A0}} = \frac{\Gamma\sinh b_1\zeta + \sinh b_1(1-\zeta)}{\sinh b_1}} \tag{42}
$$

3d.

Now, assume that there is no chemical reaction. Since $k_1''' = 0$, the governing equation is now:

$$-\mathcal{D}_{AB}\frac{d^2c_A}{dz^2} = 0$$

and subjected to the same boundary conditions. Solve the differential equation, and show that the solution without chemical reaction is:

$$\left(\frac{c_A}{c_{A0}}\right)_{\text{no rxn}} = \Gamma\zeta + (1-\zeta)$$

Do not just set $b_1 = 0$ in the solution given in problem 2c. All assumptions must be clearly stated, and all steps clearly explained.

Problem 3d: Solution

As mentioned previously, with no chemical reaction, $k_1''' = 0$ and the same assumptions as above, the governing equation becomes:

$$-\mathcal{D}_{AB}\frac{d^2c_A}{dz^2} = 0$$

Since $\mathcal{D}_{AB} \neq 0$:

$$\frac{d^2c_A}{dz^2} = 0 \tag{43}$$

Integrating once gives:

$$\frac{dc_A}{dz} = C_1 \tag{44}$$

and twice:

$$c_A = C_1 z + C_2 \tag{45}$$

The boundary conditions (same as above) can now be applied. At $z = 0$, $c_A = c_{A0}$:

$$\begin{aligned} c_{A0} &= C_1(0) + C_2 \\ c_{A0} &= C_2 \end{aligned} \tag{46}$$

At $z = \delta$, $c_A = c_{A\delta}$:

$$\begin{aligned} c_{A\delta} &= C_1\delta + c_{A0} \\ C_1 &= \frac{c_{A\delta} - c_{A0}}{\delta} \end{aligned} \tag{47}$$

Therefore:

$$c_A = \frac{c_{A\delta} - c_{A0}}{\delta}\, z + c_{A0} \tag{48}$$

Dividing both sides by c_{A0} and rearranging gives:

$$\frac{c_A}{c_{A0}} = \left(\frac{c_{A\delta}}{c_{A0}} - 1\right)\frac{z}{\delta} + 1 \tag{49}$$

and again, since $\Gamma = \dfrac{c_{A\delta}}{c_{A0}}$ and $\zeta = \dfrac{z}{\delta}$:

$$\frac{c_A}{c_{A0}} = (\Gamma - 1)\,\zeta + 1 \tag{50}$$

Which can be rewritten as:

$$\boxed{\left(\frac{c_A}{c_{A0}}\right)_{\text{no rxn}} = \Gamma\zeta + (1 - \zeta)} \tag{51}$$

3e.

Using Fick's law, show that the fluxes with and without reaction at $z = 0$ are:

With chemical reaction:

$$(N_{Az})_{z=0} = \frac{\mathcal{D}_{AB}\,c_{A0}}{\delta}\left(\frac{b_1 \cosh b_1 - b_1\Gamma}{\sinh b_1}\right)$$

No chemical reaction:

$$(N_{Az})_{z=0} = \frac{\mathcal{D}_{AB}\,c_{A0}}{\delta}\,(1 - \Gamma)$$

Problem 3e: Solution

Starting with the expression for the normalized molar concentration of gas A with the chemical reaction:

$$\begin{aligned} \frac{c_A}{c_{A0}} &= \frac{\Gamma \sinh b_1\zeta + \sinh b_1(1-\zeta)}{\sinh b_1} \\ c_A &= \frac{c_{A0}}{\sinh b_1}\left[\Gamma \sinh b_1\zeta + \sinh b_1(1-\zeta)\right] \end{aligned} \tag{52}$$

Differentiating with respect to ζ:

$$\frac{dc_A}{d\zeta} = \frac{c_{A0}}{\sinh b_1}\left[\Gamma b_1 \cosh b_1\zeta - b_1 \cosh b_1(1-\zeta)\right] \tag{53}$$

At $z = 0$, $\zeta = 0$, and $\cosh 0 = 1$:

$$\left.\frac{dc_A}{d\zeta}\right|_{\zeta=0} = \frac{c_{A0}}{\sinh b_1}\left(b_1\Gamma - b_1 \cosh b_1\right) \tag{54}$$

Furthermore, since:

$$\frac{d\zeta}{dz} = \frac{1}{\delta} \tag{55}$$

$$\frac{dc_A}{dz} = \frac{dc_A}{d\zeta}\,\frac{d\zeta}{dz} = \frac{1}{\delta}\,\frac{dc_A}{d\zeta} \tag{56}$$

Therefore:

$$\left.\frac{dc_A}{dz}\right|_{z=0} = \frac{c_{A0}}{\delta\,\sinh b_1}\left(b_1\Gamma - b_1\cosh b_1\right) \tag{57}$$

Now, using Fick's law:

$$\begin{aligned} N_{Az}\big|_{z=0} &= -\mathcal{D}_{AB}\left.\frac{dc_A}{dz}\right|_{z=0} \\ &= -\frac{\mathcal{D}_{AB}\,c_{A0}}{\delta\,\sinh b_1}\left(b_1\Gamma - b_1\cosh b_1\right) \end{aligned} \tag{58}$$

Which can be rewritten as:

$$\boxed{\left(N_{Az}\right)_{z=0} = \frac{\mathcal{D}_{AB}\,c_{A0}}{\delta}\left(\frac{b_1\cosh b_1 - b_1\Gamma}{\sinh b_1}\right)} \tag{59}$$

Now, using the expression for the normalized molar concentration of gas A with no chemical reaction:

$$\begin{aligned} \left(\frac{c_A}{c_{A0}}\right)_{\text{no rxn}} &= \Gamma\zeta + (1-\zeta) \\ c_A &= c_{A0}\left[\Gamma\zeta + (1-\zeta)\right] \end{aligned} \tag{60}$$

Differentiating with respect to ζ:

$$\frac{dc_A}{d\zeta} = c_{A0}\left(\Gamma - 1\right) \tag{61}$$

Again, since:

$$\frac{d\zeta}{dz} = \frac{1}{\delta} \tag{62}$$

$$\frac{dc_A}{dz} = \frac{dc_A}{d\zeta}\,\frac{d\zeta}{dz} = \frac{1}{\delta}\,\frac{dc_A}{d\zeta} \tag{63}$$

$$\frac{dc_A}{dz} = \frac{c_{A0}}{\delta}\left(\Gamma - 1\right) \tag{64}$$

Therefore, using Fick's law:

$$\begin{aligned} N_{Az}\big|_{z=0} &= -\mathcal{D}_{AB}\left.\frac{dc_A}{dz}\right|_{z=0} \\ &= -\frac{\mathcal{D}_{AB}\,c_{A0}}{\delta}\left(\Gamma - 1\right) \end{aligned} \tag{65}$$

Which can be rewritten as:

$$\boxed{(N_{Az})_{z=0,\ \text{no rxn}} = \frac{\mathcal{D}_{AB}\, c_{A0}}{\delta}\,(1-\Gamma)} \tag{66}$$

Program Listing: Problem 2

```
r'''
Christian DiPietrantonio
ME5130: Advanced Heat and Mass Transfer
Midterm Exam: Take-Home
Problem 02
'''

import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import fsolve


a = 1.0         # m
b = 1.0         # m
k = 0.8         # W/m-K
h = 10.0        # W/m^2-K
T0 = 100.0      # K
H = h / k       # 1/m
n_terms = 1000 # number of eigenvalues, enough for the sum to converge

def eigen_eq(bm):
    return bm * np.sin(bm * a) - H * np.cos(bm * a)

def eigenvalues(n):
    beta = np.zeros(n)
    for m in range(1, n + 1):
        guess = (m - 0.75) * np.pi / a      # guess in the middle of each solution interval
        beta[m - 1] = fsolve(eigen_eq, guess)[0]
    return beta


def print_eigenvalues(beta):
    print('+-----+----------------+')
    print('| m  | beta_m (1/m) |')
    print('+-----+----------------+')
    for m, bm in enumerate(beta, start=1):
        print(f'| {m:3d} | {bm:14.4f} |')
    print('+-----+----------------+')

def temperature_dist(x, y, beta):
    T = np.zeros_like(x)
    for bm in beta:
        norm_fac = 2 * (bm**2 + H**2) / (a * (bm**2 + H**2) + H)
        #Y_ratio = (bm * np.cosh(bm * (b - y)) + H * np.sinh(bm * (b - y))) / (bm * np.cosh(bm * b) + H * np.sinh(bm * b))
        Y_ratio = np.exp(-bm * y)* ((bm + H) + (bm - H) * np.exp(-2 * bm * (b - y)))/ ((bm + H) + (bm - H) * np.exp(-2 * bm * b)) #avoid overflow errors
        T += norm_fac * (T0 * np.sin(bm * a) / bm) * np.cos(bm * x) * Y_ratio
    return T

```

```python
def plot_contour(x, y, T, filename):
    isotherms = np.arange(10, 100, 10)
    column   = T[:, np.argmin(np.abs(x - 0.2))]
    spots = [(0.2, np.interp(L, column[::-1], y[::-1])) for L in isotherms]

    fig, ax = plt.subplots(figsize=(7, 6))
    cf = ax.contourf(x, y, T, levels=np.linspace(0, T.max(), 101), cmap="Reds")
    cs = ax.contour(x, y, T, levels=isotherms, colors="black", linewidths=0.8)
    for label in ax.clabel(cs, fmt="%.0f K", fontsize=9, manual=spots):
        label.set_bbox(dict(facecolor="white", edgecolor="none", pad=1))
    fig.colorbar(cf, ax=ax, label="$T$ (K)", ticks=np.arange(0, 101, 10))
    ax.set_xlabel("$x$ (m)")
    ax.set_ylabel("$y$ (m)")
    ax.set_title("Temperature Distribution $T(x, y)$")
    ax.set_aspect("equal")
    fig.savefig(filename, dpi=300, bbox_inches="tight")

def main():
    beta = eigenvalues(n_terms)

    # 2a: print the first ten eigenvalues
    print_eigenvalues(beta[:10])

    # 2b: create contour plot
    x = np.linspace(0, a, 501)
    y = np.linspace(0, b, 501)
    xx, yy = np.meshgrid(x, y)
    T = temperature_dist(xx, yy, beta)
    plot_contour(x, y, T, "DiPietrantonio_ME5130_Midterm_P2.png")


if __name__ == "__main__":
    main()
```